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RestSpot: A Crowdsourced Restroom Information and Accessibility App

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Abstract

Finding a suitable public restroom is a common but poorly supported need, especially for people who are traveling, in unfamiliar buildings, or have specific requirements. Restrooms exist in public buildings, businesses, parks, and transportation facilities, but information about whether a given facility is open to the public, accessible under the Americans with Disabilities Act (ADA), equipped with a baby changing table, family or gender-neutral, or reasonably clean is rarely available before a person arrives. General-purpose mapping tools can point users toward the nearest building but say little about the restroom inside it. The result is that parents with infants, people with disabilities, and others with needs often must guess or search on site. There oftentimes is not enough readily available information to make a quick decision about the best restroom nearby.

The objective of this project is to design and develop RestSpot, a crowdsourced mobile and web application that lets users find restrooms that meet their specific needs, not just the closest one. RestSpot will provide an interactive map with location-based search, filtering accessibility, amenities, access restrictions, and condition, and the ability for users to add new locations, submit ratings and reviews, and update outdated information. A supporting backend and database will store and manage this user-generated data, so the platform stays current as the community contributes. Developed in collaboration with Dr. Suman Mitra of the University of Arkansas Department of Civil Engineering, RestSpot aims to deliver a functional prototype showing how crowdsourced information and location services can make public restroom access more predictable, inclusive, and accessible.

1.0 Problem

Public restrooms in the United States are limited and hard to find. One estimate puts the national average at about eight public restrooms per 100,000 people, with fewer than three in Arkansas [1]. Most people therefore rely on restrooms inside businesses, transit stations, and other buildings. These spaces were never designed to serve the general public, and they can restrict access at any time. Before arriving, an individual usually cannot tell whether a given restroom is open to the public, accessible, equipped with a baby changing table, family or gender-neutral, or reasonably clean, resulting in dissatisfaction during the already uncomfortable situation. A restroom's hours of availability are prone to change without prior notice, its doors can be accidentally locked, and it should be noted that some establishments only permit paying customers to access its restrooms. Mapping services provide information regarding where a building is located; however, such services do not include what its restrooms have to offer. A destination that appears easy to access on a map might end at a locked door.

This lack of information matters most for people with specific needs. More than one in four U.S. adults report a disability [2], and over 33 million Americans have overactive bladders [3]. Parents with infants and other individuals in need of a single occupation or gender-neutral restroom face the same problem. For these groups, the closest restroom is not necessarily a usable one, and, without reliable information, they are left guessing. Arriving at the wrong restroom can result in a stressful search, an unsafe situation, or forfeiting the outing altogether. Many individuals respond by planning their day around restroom access or avoiding public spaces entirely. A person with an accessibility need cannot simply attempt to access the next adjacent building, and a parent carrying an infant cannot afford repeated misses. The problem is not that restrooms are absent everywhere. Rather, the concern is that the details that people need to judge whether a restroom will work for them are missing at the moment they have to decide.

2.0 Objective

The objective of this project is to design and develop RestSpot, a crowdsourced mobile and web application that helps users of the platform to locate restrooms that meet their specific needs. The application will include an interactive map with a location-based search while also allowing its users to filter by attributes such as accessibility, amenities, public availability, and condition. The application will also allow a dedicated user to add or update restroom information through ratings, reviews, and new submissions. A supporting backend database will store and manage data regarding each restroom. The final deliverable of the project is a working, published application available for public use.

3.0 Background

3.1 Key Concepts

Crowdsourcing. Crowdsourcing is a term that refers to the construction and maintenance of a dataset through contributions from a large group of individuals rather than a central authority [8]. This particular approach allows a small team of people to cover more ground than they would be able to if they were instead collecting data individually, especially if the group is spread across many cities and neighborhoods. RestSpot relies on users to add a new restroom, report a restroom's amenities, rate a restroom's conditions, and correct any outdated entries to the end that data quality depends on contribution volume in addition to ways of verifying or flagging inaccurate submissions. A restroom's information is often changing with hours of availability shifting, fixtures breaking, and the businesses in which they are located closing, so the dataset stays useful only if individuals continue to update it. Features such as confirmation of votes, buttons to send reports, and visible, verified dates provide users the ability to properly judge how much they can trust a restroom's listing.

Location services and mapping. Mobile devices possess the ability to find an individual's position by relying on a Global Positioning System (GPS) and a network-based location. GPS is typically most accurate outdoors while Wi-Fi and cellular signals aid its mapping when the device is indoors. Mapping services such as Google Maps or OpenStreetMap provide map rendering, geocoding, and directions through Application Programming Interfaces (APIs). Geocoding can involve converting a street address into coordinates, which allows users to add a restroom to the dataset by searching for a building rather than manually dropping a pin on a map. RestSpot uses these coordinates to show restrooms on an interactive map and to sort or filter results by distance. The choice of the mapping service can also affect cost and how much the map can be customized.

Mobile and web application frameworks. Frameworks such as React Native, Flutter, or native Android and iOS operating systems allow a single team to build the client that is used by an individual. The client is the part of the product that people see on their devices and includes the interactive map, the search filters, and the forms for adding a new restroom or reviewing a restroom's information. Cross-platform frameworks reduce duplicated work by allowing a wider variety of user interfaces to interact with the dataset in comparison to multiple single-platform frameworks, which require multiple copies of information. The tradeoff between the two styles is that native development can offer tighter control over performance and device features such as location, while a cross-platform approach lets a small team ship to both Android and iOS faster and keep both versions consistent.

Backend and database. A backend service manages user accounts, submissions, and queries without direct interaction from the users. It also enforces various rules on the

platform, such as limiting how often a user can submit and ensuring that each new entry contains information in all the required fields before the information is saved. A database stores restroom records, attributes, ratings, and reviews. Each record can hold a restroom's location, a set of amenities such as the restroom being accessible or including a baby-changing table, and a history of changes so that inaccurate edits can be revised. Location-based search requires geospatial queries, meaning the database can efficiently find records within a given distance of a point. Without spatial indexing, these searches' times of completion slow down as the number of restrooms increases. As a result, the choice of the database is an important factor when regarding how to keep the computation time of results small.

3.2 Related Work

There already exist several applications that address certain aspects of the presented issue. Refuge Restrooms [4] is an open-source, crowdsourced platform focused on safe restrooms for transgender, intersex, and gender-nonconforming individuals. Users of the platform can search for restrooms by location, add new listings, and post comments and rate existing restrooms on the platform. The focus of Refuge Restrooms is narrow by design, so it is not intended as a general restroom directory. Flush Toilet Finder [5], another application, has a much larger database of about 200,000 restrooms worldwide and allows its users to add, rate, and report restrooms. The platform's listings carry only a few attributes, such as disabled access, fees, or key requirements. Google Maps [6] provides information regarding whether a place has a wheelchair-accessible restroom, but this is only given as a single attribute of a business rather than a listing of its own, and the service does not cover other attributes, such as changing tables, family restrooms, public access, or cleanliness.

The proposed project differs from the previous applications by treating the restroom as the primary record rather than an addition to a larger set of information. The application combines a broad set of filterable attributes that cover accessibility, family and gender-neutral options, changing tables, access restrictions, hours of availability, and cleanliness conditions in a single, crowdsourced platform aimed at all users rather than a single group.

4.0 Design

4.1 Use Cases

RestSpot has three user roles: *Guest*, *Registered User*, and *Administrator*:

Guest: The *Guest* has the ability to view nearby restrooms, search by location, filter results, view restroom details, get directions, and create an account. After creating an account, the *Guest* is promoted to a *Registered User*.

Registered User: A *Registered User* can perform all *Guest* use cases except for the ability to create a new account. The *Registered User* can also manage an account, add a new restroom, edit a restroom's information, rate and review restrooms, report a problem, and save favorites. The *Registered User* can also delete the user's associated account.

Administrator: There is no direct way for a *Guest* or *Registered User* to become an *Administrator* as this role is reserved for those involved in the platform's production and maintenance. The *Administrator* can review reports, merge duplicates, and moderate *Guests* and *Registered Users* with the authority to restrict access to certain or all aspects of the platform for users that are found to be operating in an unseemly manner.

4.2 Requirements

4.2.1 Functional Requirements

The system must provide location detection, proximity search, filtering, restroom records, ratings and reviews, user accounts, edit history, duplicate detection, problem reporting, and user moderation.

Location and search. Location detection uses the device's GPS coordinates in decimal degrees, with latitude from −90° to 90° and longitude from −180° to 180°. If location access is denied, the user can enter an address, which is geocoded. Proximity search returns restrooms within a user-selected radius of 0.1 miles to 25 miles that defaults to 2 miles and is sorted by distance. Filtering takes boolean inputs for ADA accessibility, changing table availability, family restroom classification, gender-neutral restroom classification, and public accessibility. It also offers an "open now" option that compares stored hours to the device's local time, and a minimum cleanliness rating from 1 to 5.

Restroom data. Each restroom record stores the following fields:

- A facility name of 1 to 100 characters.
- Coordinates.
- An access type of public, customers only, or restricted.
- A restroom type of single-occupancy, multi-stall, or family.
- Boolean amenity attributes.
- Hours for each day of the week in 24-hour HH:MM format, marked as 24 hours, or unknown.
- Optional notes of up to 500 characters.

Each edit records the ID of the user that created the entry, a timestamp of the time the entry was uploaded or updated, and the changed fields. When a user adds a restroom, duplicate detection warns him or her if an existing listing with a similar name is within 50 meters.

User input and moderation. Ratings and reviews consist of a cleanliness rating and an overall rating. Each of these values is to be an integer in the inclusive range of 1 to 5. The operation will also include an optional comment of up to 500 characters. Averages are displayed with an accuracy of up to one digit past the decimal place. User accounts require a valid email address as well as a password of at least 8 characters. Only authenticated users can contribute to the database. Reports indicating a problem with a restroom must include a reason (the restroom is closed, the listing contains incorrect information, the listing is a duplicate, or the listing is inappropriate) and an optional comment of up to 500 characters. These reports are sent to a queue for administrative review. Moderation allows administrators to edit or remove listings and reviews, merge duplicates, and suspend accounts.

4.2.2 Interface Requirements

The client communicates with the backend over Hypertext Transfer Protocol Secure (HTTPS) using a Representational State Transfer API (REST) based on JavaScript Object Notation (JSON), or through the backend provider's Software Development Kit (SDK) if the team uses a managed service. A third-party mapping API provides map display and geocoding. Directions are handled by deep linking to the device's default maps app. The client uses the operating system's location services, which require user permission. Because the app is built around accessibility, the user interface should follow Web Content Accessibility Guidelines (WCAG) 2.1 AA, including screen reader labels, sufficient color contrast, and scalable text.

4.2.3 Performance Requirements

On a typical Long-Term Evolution (LTE) broadband connection, the map and nearby results load within 3 seconds of opening the app. Changing a filter updates results within 1 second. Proximity queries are completed on the server within 500 milliseconds for a dataset of at least 10,000 restrooms. New submissions and edits are visible to other users within 5 seconds of saving. The prototype supports at least 100 concurrent users. No formal uptime target applies at the prototype stage.

5.0 Key Personnel

Luke Fletcher – Fletcher is a senior Computer Engineering major in the Electrical Engineering and Computer Science Department at the University of Arkansas. He has participated in cybersecurity testing internships with the United States Air Force Civilian Service at Eglin AFB.

Lane Harper – Harper is a senior Computer Science major in the Electrical Engineering and Computer Science Department at the University of Arkansas. Harper is also studying for a concentration in Cybersecurity as well as a minor in Mathematics in the Department of Mathematical Sciences at the University of Arkansas. Harper completed a brief internship at Hazen School District in Hazen, Arkansas, in which he aided in the field of Information Technology.

Alex Hayek – Hayek is a senior Computer Science major with a concentration in Cybersecurity in the Electrical Engineering and Computer Science Department at the University of Arkansas.

Clark Kirby – Kirby is a senior Computer Science major in the Electrical Engineering and Computer Science Department at the University of Arkansas. He has completed software engineering internships at Walmart Global Tech and at Honeywell at the Kansas City National Security Campus.

Eli McCrary – McCrary is a senior Computer Science major with a concentration in Cybersecurity in the Electrical Engineering and Computer Science Department at the University of Arkansas. He has completed a software internship at Adamas Energy in Dallas, Texas.

Will Whitlow – Whitlow is a senior Computer Science major in the Electrical Engineering and Computer Science Department at the University of Arkansas.

Dr. Suman Mitra – Dr. Mitra is an associate professor in the Department of Civil Engineering at the University of Arkansas. His research activity involves aspects of transportation and travel [7]. Dr. Mitra is the sole sponsor of the project, providing the initial idea of RestSpot and supervising the application's production.

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